BrightNest

Grounded after-hours help for parents

COMP6708 Mini Demo · **Group 15**

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seanfontaine.dev/COMP6708_A3_Demo

One sentence

1

A learning-centre assistant that

answers from centre truth,
acts with tools,
and hands off when humans must.

Training-free · RAG · hybrid routing · live on the public web

The problem

1

Imagine you run a tutoring centre.

- Parents message after hours: typhoons, fees, seats, trials, booking
- Staff cannot live on WhatsApp 24/7
- A wrong fee or hours answer costs trust — fast

So you hired Group 15

A couple of weeks later we built **BrightNest**: grounded answers, structured actions, honest hand-offs.

BrightNest

Not a chatbot that invents tuition from thin air.

Technique: training-free RAG + tools

1

Why training-free? Centre docs change weekly ⇒ fine-tuning is the wrong loop.

RAG (self-built)

1. Chunk markdown KB (`content/`)
2. Embed with DashScope
3. Store / query in Chroma
4. Lexical rerank → top-k
5. Generate with Qwen + citations

Retrieval score $s(q, c) = \mathrm{sim}(E(q), E(c))$ · inspector shows rank, s, full chunk.

Hybrid router

- Scores are **fixed detector awards**, not RAG sim
- Soft FAQ: grounded compose (card > KB)
- Hard FSM (book / cancel / escalate) bypasses retrieve
- Cheap arbiter only if low confidence

card privileged **KB may enrich**

score ledger in inspector

Three answer tiers

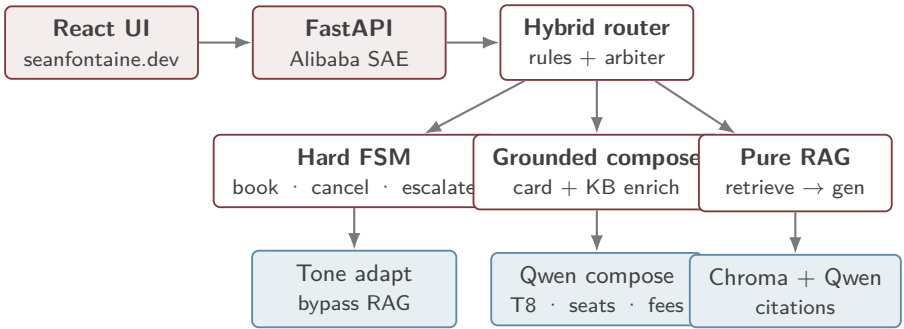
1

Tier	Examples	RAG role
Hard FSM / hand-off	Book / cancel dialogue, escalate, refuse	Bypass
Grounded compose	Typhoon+HKO, seats, fees, allergy	Live doc + KB enrich
Pure RAG	Private 1:1, multi-topic narrative	Full generate

Intent score e.g. `fees=0.90` means “fee regex fired” — open the inspector ledger to see every award / blend / clear.

Architecture

1



Inspector: match cue · score ledger · grounding mode · enrich chunks · session memory

What you will see live

2

Parent chat (WhatsApp-shaped)

- Messaging surface + probes
- Chat only — no booking form / Book buttons

System inspector

- Router decision + score ledger
- Grounding mode · RAG expand
- Class register · session memory

Staff desk

- SheetPro follow-ups (gaps / assessment)
- MailBox urgent inbox

5-minute rule

Slides cue. We **operate** the system.

Open the live demo fullscreen. Reset once.

Demo beats (cue card)

2

Beat	Show	Point at
1	Typhoon	Score 1.0 · live/policy privileged
2a	Phonics fee	fees=0.90 ledger · grounded compose
2b	RAG · Private 1:1	Pure retrieve · expand chunk
3	Seats → Book Jason	Chat seats · register · assessment→SheetPro
4	Scuba (unknown)	Follow-up **SheetPro** row
5	Safety allergy call	**MailBox** urgent mail

Beats 1→5 without reading bullets. Narrate the inspector.

Impact & why it works

3

Impact

- 24/7 triage without inventing policy
- Tools for hours / fees / seats
- RAG for programme narrative
- Humans when it matters

Creativity

- Hybrid router + confidence-gated review
- Staff desk = Sheets + Gmail clones

Correctness cues

Visible rule map + decision path

Citations with retrieval scores

Urgent safety inbox + email draft

Off-topic refuse (no fake todo)

Evaluation

Live path correctness · visible RAG · latency feel.
No formal user study required.

Thank you

3 Impact

Grounded answers. Structured actions. Honest hand-offs.

RAG · tools · staff desk

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Q&A

Thank you for listening!
Your feedback will be highly appreciated!